Citi Bike Data Analysis

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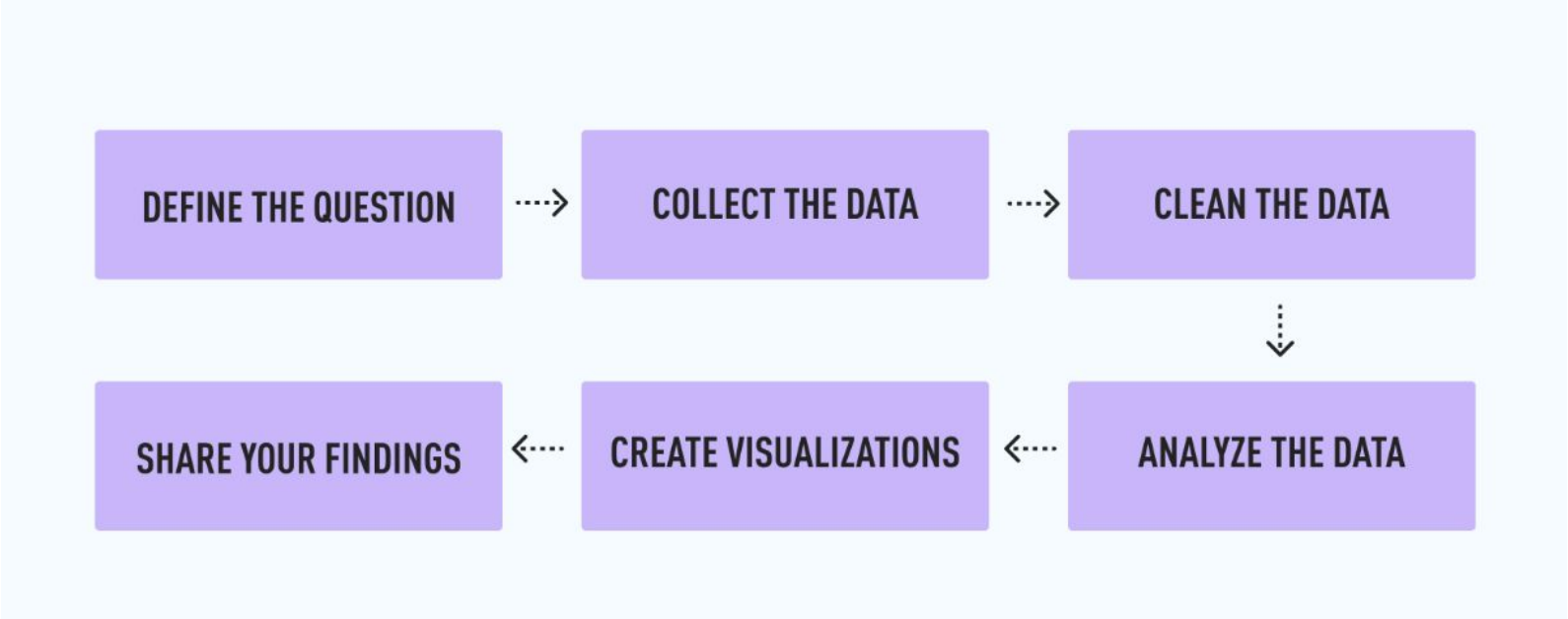
Project Goal:

- *To better understand the behavior of Citi Bike's customer base (both one-time users and subscribers) and how they use Citi Bikes*

- *This will help us to:*
  - *Identify where more bikes should be installed*
  - *Create targeted marketing campaigns that will appeal to different customer segments*

Data Analytics Approach



Business Questions

1. *How does the average trip duration vary across different age groups?*
2. *Which age group rents the most bikes?*
3. *How does bike rental vary across the two user groups (one-time users vs long-term subscribers) on different days of the week?*
4. *Does user age impact the average bike trip duration?*
5. *What are the most popular pick-up locations across the city for NY Citi Bike rental?*

Dataset Overview

Dataset

- ~16,800 Citi Bike trips
- NYC | Jan–Mar
- Each row = one trip

Core Fields

- Age
- Bike ID
- Start & End Time
- Start & End Station
- User Type
- Temperature

Derived for Analysis

- Trip Duration
- Age Groups
- Weekday / Month / Season

Descriptive statistics

Statistic	Trip Duration	Age
Mean	9	42
Median	5	39
Minimum	1	22
Maximum	3,693	90

Data Cleaning

→ Handling missing values

→ Removing duplicate records

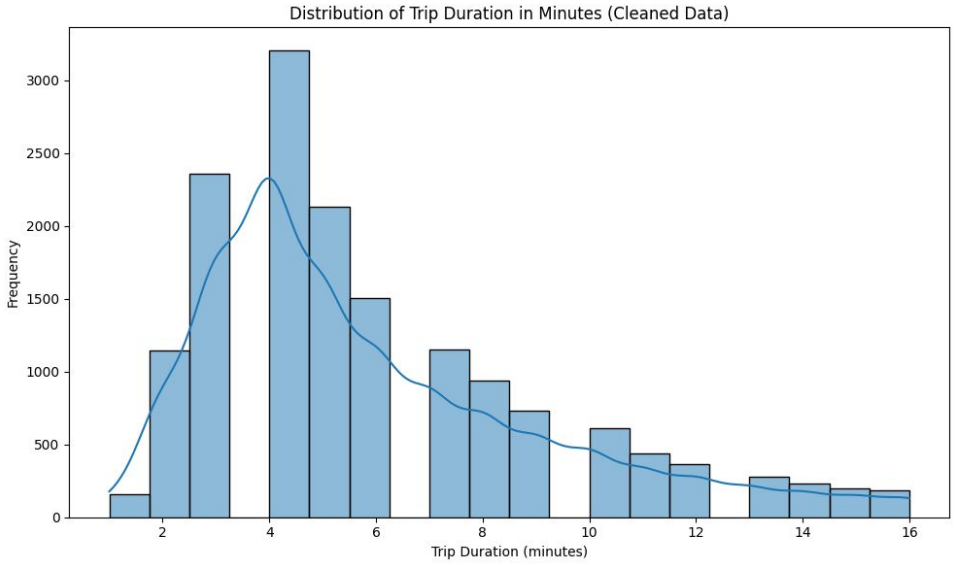
→ Fixing inconsistent entries

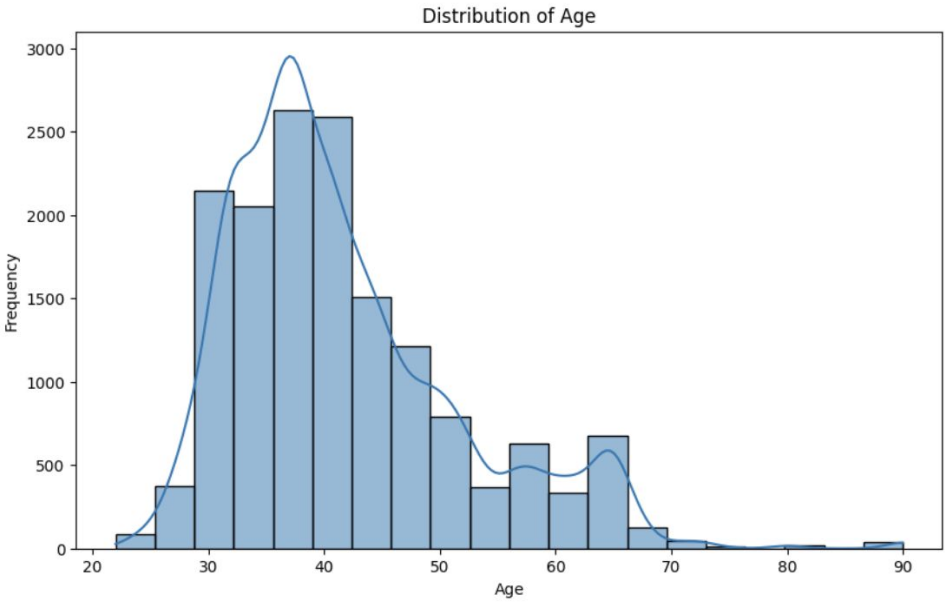
→ Standardizing naming formats

→ Removing extreme outliers

→ Clean, reliable data ready for analysis

Understanding Data Distribution





Trip duration is right-skewed, with most trips being short and a long tail of longer trips

Majority of riders fall between ages ~25–45, with fewer users at higher age ranges

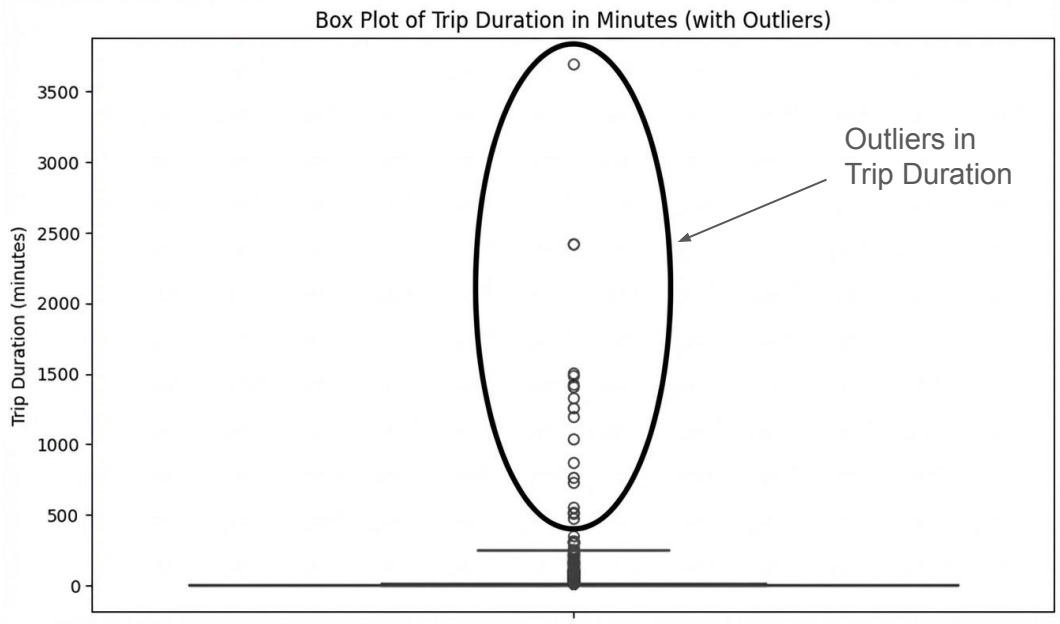
Outlier Detection in the Trip Duration

Q1 (25th percentile): 4.00 mins
Q3 (75th percentile): 9.00 mins

IQR: 5.00 minutes

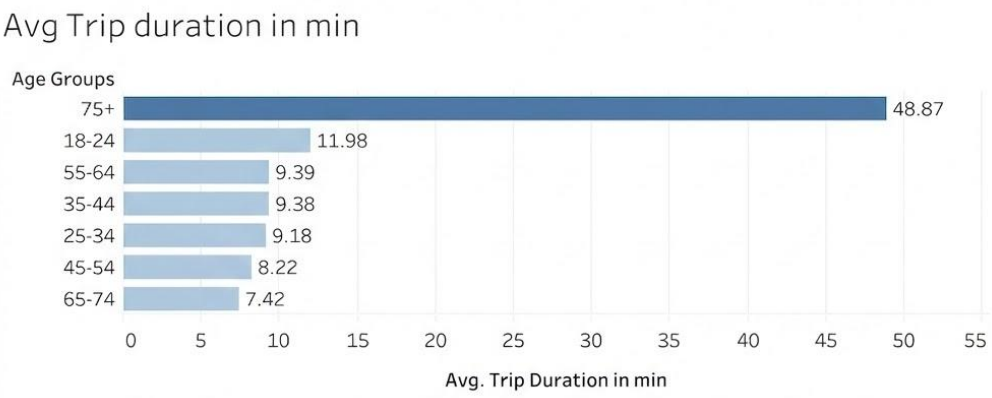
Lower Bound for Outliers: -3.50 mins
Upper Bound for Outliers: 16.50 mins

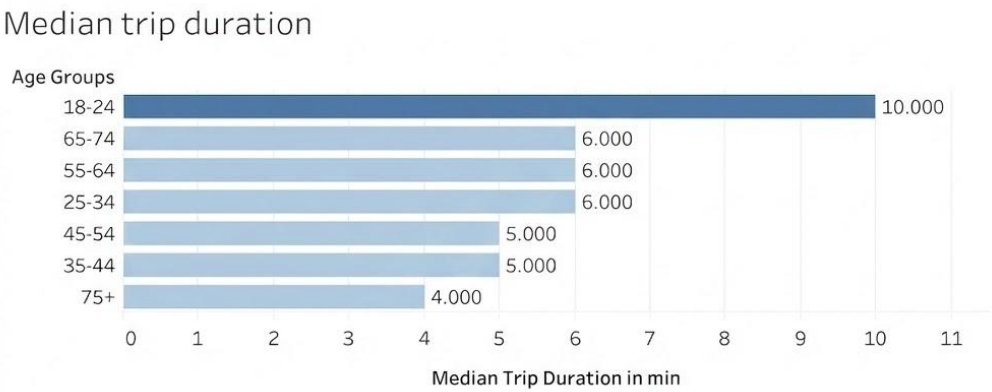
Number of outliers identified: **1218**
Percentage of outliers: **7.23%**



Impact of Outliers on Trip Duration Analysis

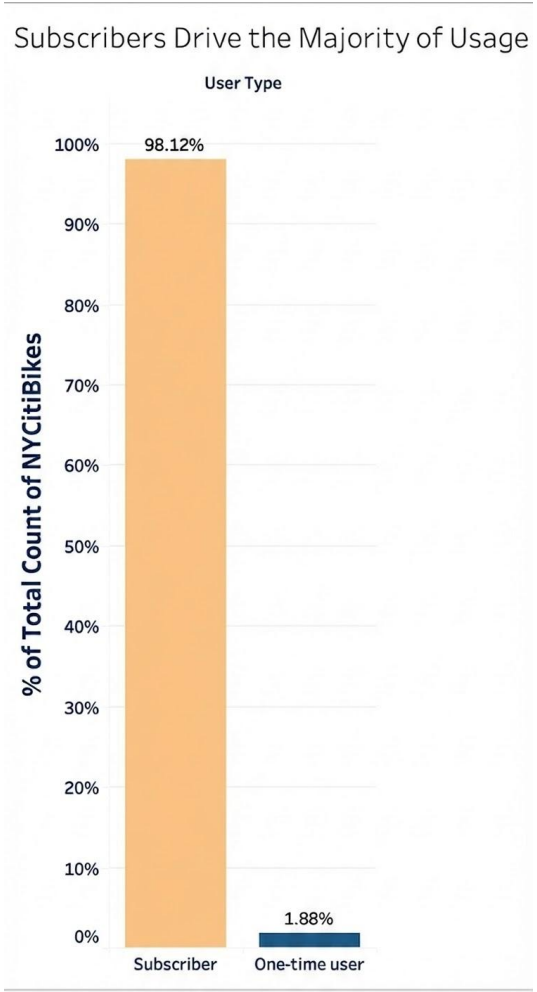
- Average can be misleading
  - The 75+ age group shows a significantly higher **average** trip duration compared to other age groups.
- Driven by outliers and small sample size
  - This unusually high average is caused by **a small number of very long trips** and **fewer overall rides** in the 75+ group.
- Median gives a more realistic view





Most Trips Come from Subscribers

- Subscribers account for **~98% of total trips**, indicating strong recurring usage
- One-time users represent a **very small share (~2%)** of overall activity
- Usage is driven by **habitual, repeat behavior rather than occasional riders**

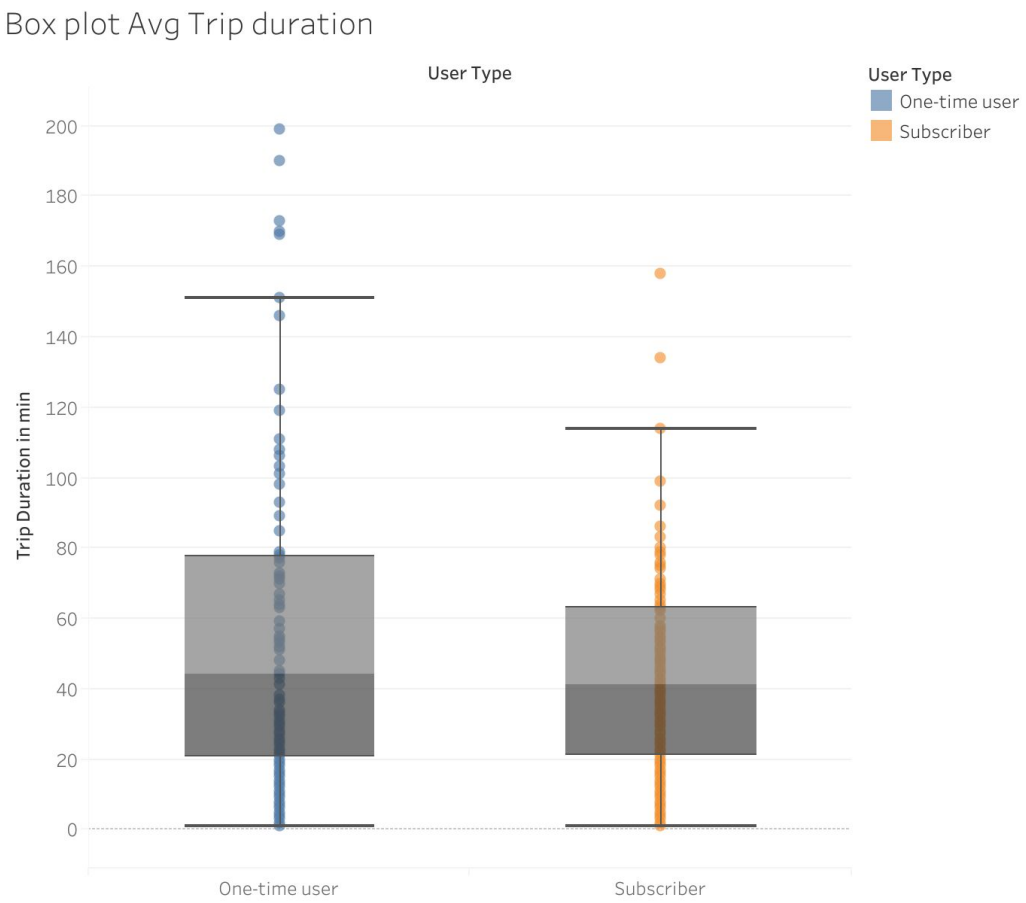


Trip Duration by User Type

One-time users show **higher variability** in trip duration.

Subscribers have **more consistent trip durations**

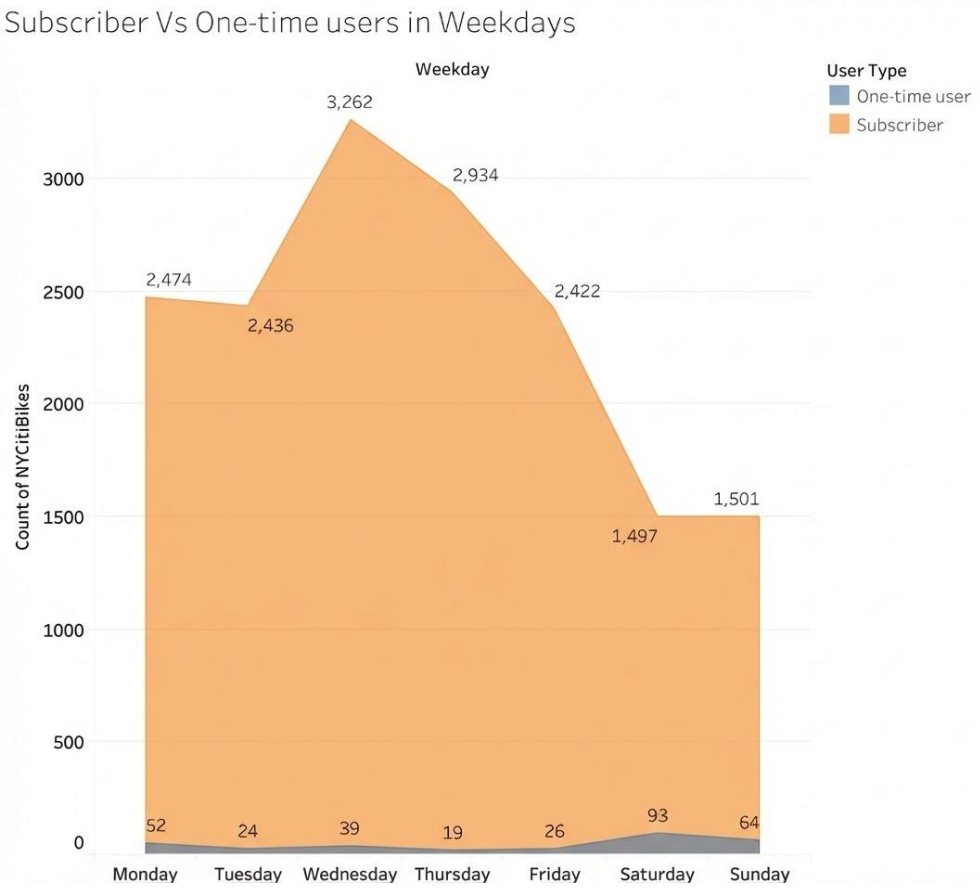
Outliers exist for both groups and influence averages



Weekday Usage Patterns by User Type

- Subscriber usage is consistently high throughout weekdays, peaking mid-week **(Wednesday–Thursday).**

- One-time users contribute a very small share overall, with relatively **higher activity on weekends.**

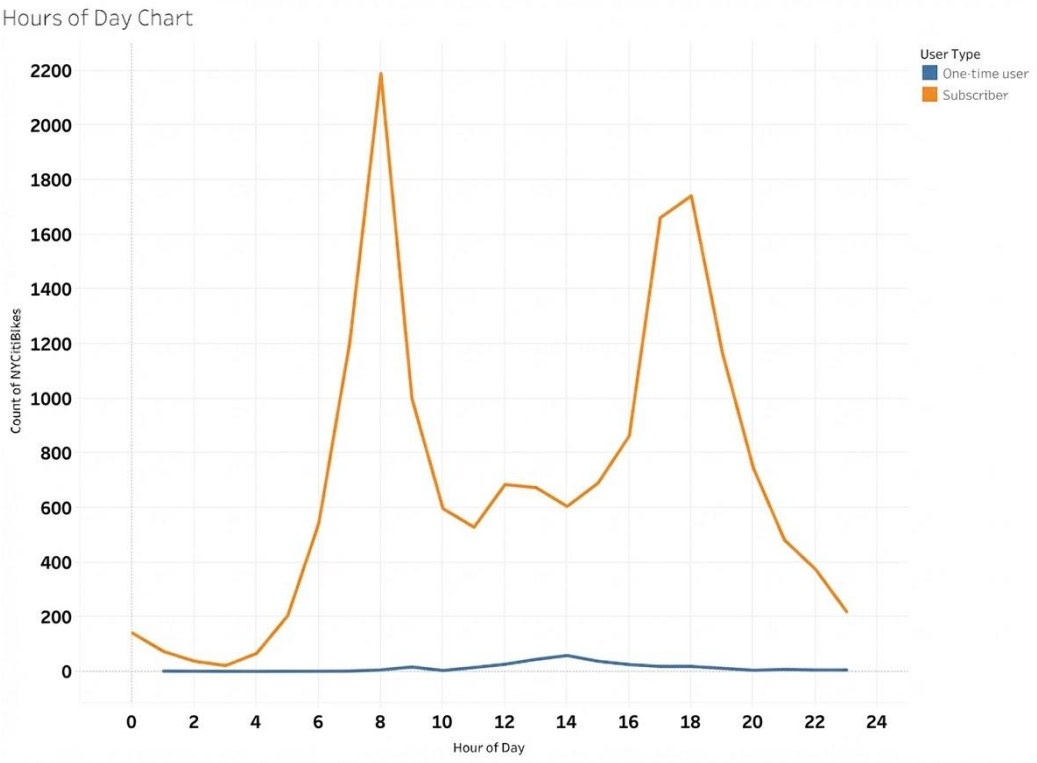


When Do Users Ride?

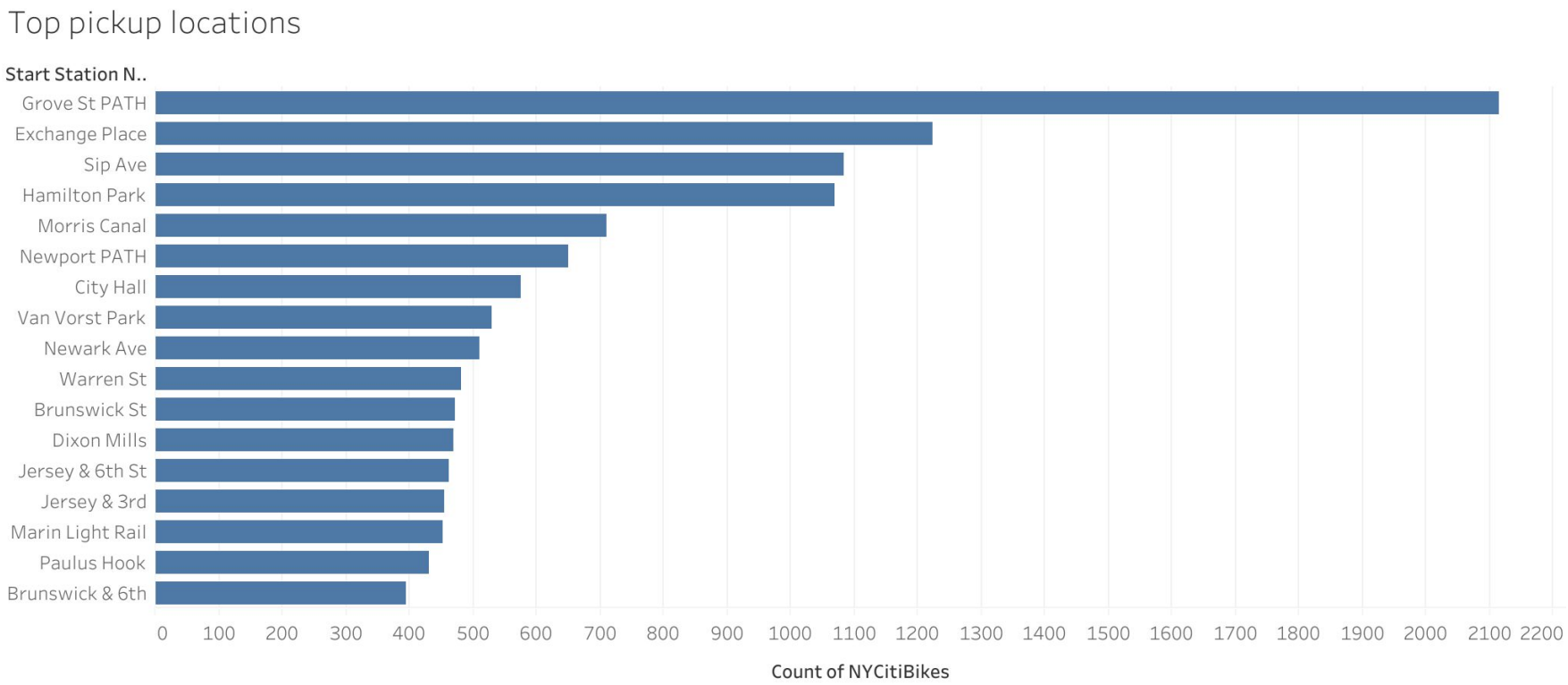
- Subscriber usage shows two clear peaks: **morning (around 8 AM) and evening (around 5–6 PM).**

- Midday usage remains steady but lower, suggesting non-commute or flexible trips.

- One-time users show much lower and flatter usage throughout the day.



High-Demand Pickup Locations Across NYC



Business Questions and Answers

1. **How does the trip duration vary across different age groups?**
   The age group of 18-24 has the highest median trip duration.

2. **Which age group rents the most bikes?**
   Riders in the mid-20s to mid-40s age range account for the highest number of bike rentals.

3. **How does bike rental vary across the two user groups on different days of the week?**
   Subscribers consistently dominate bike usage throughout the week, especially on weekdays, while one-time users show relatively higher activity on weekends.

4. **What are the most popular pick-up locations across the city for NY Citi Bike rental?**
   The most popular pick-up locations are concentrated at Grove path, Exchange place etc

Thank you!